

Shock Transmission to USD/VND Exchange Rate Volatility: Evidence from EGARCH-X and SVAR Models

Pham Khanh Linh

DSEB-MFE, NEU, Course: Time Series, Email: 11230560@st.neu.edu.com

Abstract—This study examines shock transmission to USD/VND exchange rate volatility using daily data from 2010 to 2026. USD/VND log returns are modelled together with global and domestic financial shock variables, including VIX changes, U.S. 10-year Treasury yield changes, DXY returns, oil returns, and VN-Index returns. The empirical framework combines preliminary time-series diagnostics, ARMA mean filtering, GARCH-family volatility models, EGARCH-X variance specifications, HAC shock-volatility regressions, and supplementary weekly SVAR analysis. The results show that USD/VND returns exhibit strong volatility clustering, conditional heteroskedasticity, fat tails, and persistent conditional volatility. Among the baseline volatility models, the ARMA(3,2)-EGARCH(1,1)-t specification performs best in capturing the conditional variance process. The EGARCH-X evidence indicates that the February 2010 devaluation episode improves model fit, while richer shock specifications provide selective evidence that lagged absolute oil and DXY shocks are associated with higher volatility. HAC regressions further suggest that absolute shock magnitudes explain volatility better than signed shock directions. Weekly SVAR evidence is supportive but limited, showing predictive relevance mainly for U.S. interest-rate shocks in returns and oil shocks in volatility. Overall, the strongest evidence concerns volatility persistence, fat-tailed innovations, and major exchange-rate policy events, while evidence for broader global shock transmission is selective rather than uniform.

Index Terms—Time Series, USD/VND; exchange rate volatility; GARCH-type models; global shocks; Vietnam; financial risk.

I. INTRODUCTION

Exchange rate volatility is an important source of macro-financial risk in developing economies. Fluctuations in the exchange rate can affect inflation, trade competitiveness, external debt valuation, capital flows, and overall financial stability. These effects are particularly relevant for economies with less developed financial markets because firms and investors may have limited access to hedging instruments, while domestic financial systems may be less able to absorb external shocks. As a result, global financial disturbances can be transmitted more strongly into domestic currency markets.

Vietnam provides a relevant case for studying exchange rate volatility because the USD/VND exchange rate operates under a managed exchange rate environment rather than a fully floating regime. In such a setting, exchange rate movements may appear relatively stable during normal periods, but large adjustments can still occur during episodes of policy pressure, global financial stress, or changes in international dollar conditions. Therefore, the volatility of USD/VND returns may not be explained only by standard market fluctuations. It may also reflect the interaction between volatility persistence, exchange-rate

management, global financial shocks, and domestic financial-market conditions.

This study examines the transmission of global and domestic financial shocks to USD/VND exchange rate volatility using daily data from 2010 to 2026. The dependent variable is the daily log return of the USD/VND exchange rate. The global shock variables include changes in the CBOE Volatility Index (VIX), changes in the U.S. 10-year Treasury yield, returns on the U.S. Dollar Index (DXY), and returns on WTI crude oil prices. These variables capture global risk aversion, international interest-rate conditions, broad U.S. dollar movements, and commodity-market shocks. The domestic financial variable is the VN-Index return, which is used as a proxy for domestic equity-market conditions. Major exchange-rate and crisis episodes are also controlled for using event dummy variables.

The empirical strategy proceeds in several stages. First, preliminary time-series diagnostics are used to examine stationarity, serial dependence, volatility clustering, conditional heteroskedasticity, non-normality, and multicollinearity among the shock variables. Second, an ARMA conditional mean model is selected to filter short-run dependence in USD/VND returns. Third, several GARCH-family models are compared under Student-(t) innovations in order to identify the most suitable baseline volatility specification. Fourth, the selected volatility model is extended into an EGARCH-X framework, where global and domestic shock variables enter the conditional variance equation directly. Fifth, HAC shock-volatility regressions are estimated as supplementary reduced-form checks. Finally, weekly VAR/SVAR models are used to examine dynamic shock transmission and predictive relationships between global shocks, domestic financial conditions, USD/VND returns, and selected conditional volatility.

This paper addresses four research questions. First, do USD/VND returns exhibit volatility clustering, conditional heteroskedasticity, and fat-tailed innovations? Second, which GARCH-family specification best captures USD/VND conditional volatility after filtering short-run mean dependence? Third, do global and domestic financial shocks explain movements in USD/VND conditional volatility? Fourth, are shock magnitudes more informative than signed shock directions in explaining exchange-rate volatility?

The study contributes to the literature in three ways. First, it extends the domestic-global volatility framework to the Vietnamese foreign exchange market, where exchange-rate management may affect both the level and the timing of

volatility transmission. Second, it distinguishes between signed shocks and absolute shocks, allowing the analysis to test whether the direction or the magnitude of financial disturbances matters more for USD/VND volatility. Third, it combines daily EGARCH-X volatility modelling with HAC regressions and weekly SVAR evidence, thereby separating the main conditional-volatility results from supplementary reduced-form and dynamic transmission checks.

The remainder of the paper is organized as follows. Section II reviews the related literature on exchange rate volatility, GARCH-type models, global and domestic shock transmission, and Vietnamese exchange-rate studies. Section III describes the data, variable construction, diagnostic tests, and empirical methodology. Section IV reports the empirical results. Section V discusses the main findings and concludes.

II. LITERATURE REVIEW

Exchange rate volatility has been widely studied in international finance because it affects trade flows, investment decisions, inflation dynamics, and financial stability. In developing economies, exchange rate movements may be especially important because financial markets are often less complete and firms may have weaker access to external finance and currency hedging instruments. Aghion et al. [?] argue that the effect of exchange rate volatility depends on the level of financial development. In financially underdeveloped economies, exchange rate fluctuations can have stronger real effects because financial frictions amplify the impact of external disturbances. This theoretical argument supports the view that exchange rate volatility in developing economies should be analyzed not only as a statistical feature of currency returns, but also as a possible channel through which global financial shocks affect domestic macro-financial stability.

From a methodological perspective, the standard approach to modelling financial-market volatility begins with ARCH and GARCH-type models. Engle [?] introduced the ARCH model to capture time-varying conditional variance, while Bollerslev [?] extended this framework into the GARCH model by allowing current volatility to depend on both past squared innovations and past conditional variance. These models are suitable for exchange-rate returns because currency returns commonly display volatility clustering, conditional heteroskedasticity, and heavy-tailed distributions. Under volatility clustering, periods of large exchange-rate movements tend to be followed by further large movements, while tranquil periods tend to persist.

Subsequent developments in volatility modelling allow for richer conditional-variance dynamics. Nelson [?] proposed the EGARCH model, which models the logarithm of conditional variance and allows volatility to respond asymmetrically to standardized shocks. Glosten et al. [?] developed the GJR-GARCH model, another common asymmetric volatility specification. These extensions are useful because financial markets may react differently to positive and negative innovations. However, in a managed exchange-rate environment, the magnitude of shocks may be more important than their sign if exchange-rate adjustment is influenced by policy smoothing, trading

bands, or administrative intervention. Therefore, comparing symmetric GARCH, EGARCH, and GJR-GARCH models provides a useful basis for selecting the most appropriate volatility specification for USD/VND returns.

Recent studies have moved beyond estimating volatility persistence and have focused on identifying the sources of exchange-rate volatility. Ozkaya and Altun [?] examine domestic and global causes of Turkish lira exchange-rate volatility using a GARCH framework and Lyapunov exponents. Their study separates global financial variables, such as the VIX and the dollar index, from domestic variables, such as CDS and inflation expectation differentials. Their results show that both global risk aversion and domestic financial risk can affect exchange-rate volatility. This provides an important reference for the present study because it motivates the separation between external financial shocks and domestic market conditions.

External financial conditions are also emphasized in studies of emerging and developing Asian markets. Caporale et al. [?] examine the effect of equity and bond portfolio inflows on exchange-rate volatility in seven Asian developing and emerging economies. Using GARCH models and Markov-switching specifications, they show that international portfolio flows can be an important source of currency volatility. This finding supports the broader view that exchange rates in emerging markets are sensitive to global liquidity, investor sentiment, and international capital allocation. For Vietnam, this implies that variables such as global risk aversion, U.S. interest-rate conditions, U.S. dollar strength, and commodity-price movements may contain relevant information for USD/VND volatility.

The role of the U.S. dollar and U.S. financial conditions is particularly relevant for USD/VND. Movements in the U.S. Dollar Index capture broad changes in dollar strength against major currencies, while U.S. Treasury yield changes reflect shifts in global interest-rate conditions and the attractiveness of dollar-denominated assets. Increases in global risk aversion, proxied by the VIX, may also affect emerging-market currencies through capital-flow pressure and safe-haven demand for U.S. dollar assets. Oil-price shocks can influence exchange-rate pressure through trade-balance effects, inflation expectations, and broader commodity-market uncertainty. These channels justify the inclusion of VIX changes, U.S. Treasury yield changes, DXY returns, and oil returns as global shock variables in the empirical model.

For Vietnam, previous studies have mostly examined the consequences of exchange-rate volatility rather than its determinants. Thuy and Thuy [?] study the impact of exchange-rate volatility on Vietnam's exports using an ARDL approach. Their focus is on how volatility affects trade performance rather than on which domestic or global shocks generate exchange-rate volatility. Similarly, Huynh and Hoang [?] analyze the effect of exchange-rate volatility on Vietnam's bilateral imports, using GARCH-based volatility as one explanatory measure. These studies confirm that exchange-rate volatility matters for Vietnam's external sector, but they do not directly investigate

the financial sources of USD/VND volatility.

This gap is important because Vietnam's exchange-rate regime differs from a fully floating currency market. Under a managed exchange-rate environment, observed exchange-rate volatility may reflect both market pressure and policy smoothing. Daily USD/VND returns can therefore exhibit long quiet periods interrupted by discrete adjustment episodes. This makes the Vietnamese case suitable for GARCH-family models with event controls and external shock variables. It also means that shock transmission should be interpreted carefully: insignificant signed shocks do not necessarily imply the absence of external pressure, because policy management may delay or smooth the observed exchange-rate response.

In addition to GARCH-type models, VAR and SVAR frameworks are commonly used to study dynamic shock transmission. Sims [?] introduced VAR analysis as a way to examine interdependence among macroeconomic time series without imposing overly restrictive single-equation structures. In the context of this study, weekly VAR/SVAR models are useful as supplementary evidence because they allow global and domestic variables to interact dynamically with USD/VND returns and selected conditional volatility. However, SVAR results depend on identifying assumptions and variable ordering. Therefore, they are best interpreted as dynamic supporting evidence rather than as the main source of causal inference.

Overall, the literature suggests four main implications for this study. First, exchange-rate volatility is economically important in developing economies because financial frictions can amplify the effect of shocks. Second, GARCH-family models are appropriate for USD/VND returns if the data exhibit volatility clustering, conditional heteroskedasticity, and fat-tailed innovations. Third, recent empirical studies support the distinction between global and domestic determinants of exchange-rate volatility. Fourth, because Vietnam operates under a managed exchange-rate environment, shock transmission should be analyzed with attention to event episodes, volatility persistence, and the possibility that shock magnitudes matter more than signed shock directions.

Following this literature, the present study examines USD/VND exchange-rate volatility using an ARMA-EGARCH-X framework supplemented by HAC regressions and weekly SVAR analysis. Unlike studies that focus only on the effects of exchange-rate volatility on trade, this paper investigates the determinants and transmission channels of volatility itself. The study therefore contributes to the Vietnam-focused literature by evaluating whether global risk aversion, U.S. interest-rate changes, U.S. dollar movements, oil-price shocks, and domestic equity-market conditions help explain USD/VND conditional volatility over the 2010–2026 period.

III. DATA AND METHODOLOGY

A. Data Source and Cleaning

This study uses daily time-series data to examine the transmission of global and domestic financial shocks to USD/VND exchange rate volatility. The raw dataset contains the USD/VND nominal exchange rate, the CBOE Volatility

Index (VIX), the U.S. 10-year Treasury yield, the U.S. Dollar Index (DXY), WTI crude oil prices, and the VN-Index. The final modelling period runs from 26 January 2010 to 31 December 2025 after data cleaning, return construction, rolling-volatility construction, and removal of unavailable observations.

The data are cleaned before feature construction. The date variable is converted into datetime format, observations are sorted chronologically, and the holiday observation on 1 January 2026 is excluded. Market variables are converted into numeric format. For VIX, U.S. 10-year Treasury yield, WTI oil prices, DXY, and VN-Index, missing values are forward-filled with a maximum limit of five consecutive days. VN-Index values are additionally cleaned by removing comma separators before numeric conversion. After this cleaning step, the full feature-engineered dataset contains 4,021 daily observations and no missing values in the raw market variables.

The model-ready dataset is obtained after constructing returns, changes, and rolling volatility. Because the 22-day rolling volatility requires past observations, the final model-ready sample begins on 26 January 2010. The model-ready dataset contains 4,006 observations. Some specifications involving oil returns, lagged shock variables, or selected conditional volatility use slightly fewer observations because those variables are subject to listwise deletion within the corresponding model.

B. Variable Construction

The main dependent variable is the daily USD/VND log return. Let S_t denote the USD/VND nominal exchange rate. The exchange rate return is defined as

$$r_t = 100 [\ln(S_t) - \ln(S_{t-1})]. \quad (1)$$

The exchange rate return is used instead of the exchange rate level because volatility modelling concerns the conditional variability of short-run exchange rate movements rather than the level of the exchange rate itself.

Several preliminary volatility proxies are constructed for descriptive analysis and diagnostic testing. These include the absolute return, squared return, and rolling 22-day standard deviation:

$$\begin{aligned} a_t &= |r_t|, \\ q_t &= r_t^2, \\ \sigma_t^{(22)} &= \left[\frac{1}{21} \sum_{j=0}^{21} \left(r_{t-j} - \bar{r}_t^{(22)} \right)^2 \right]^{1/2}. \end{aligned} \quad (2)$$

In the implementation, the rolling volatility is computed as a 22-day rolling sample standard deviation with a minimum of 15 available observations. These variables are not used as substitutes for GARCH-type conditional volatility. They are used only as preliminary indicators of volatility clustering and time-varying exchange rate risk.

The global financial shock variables are constructed as follows:

$$\begin{aligned}
\Delta VIX_t &= VIX_t - VIX_{t-1}, \\
\Delta US10Y_t &= US10Y_t - US10Y_{t-1}, \\
r_t^{DXY} &= 100 [\ln(DXY_t) - \ln(DXY_{t-1})], \\
r_t^{OIL} &= 100 [\ln(OIL_t) - \ln(OIL_{t-1})].
\end{aligned} \tag{3}$$

The VIX and U.S. 10-year Treasury yield are transformed into first differences because they are already index or rate variables. DXY and WTI crude oil prices are transformed into log returns because they are price-level variables.

The domestic financial-market variable is the VN-Index return:

$$r_t^{VNI} = 100 [\ln(VNINDEX_t) - \ln(VNINDEX_{t-1})]. \tag{4}$$

The VN-Index return is used as a domestic financial-market control rather than as a global shock. This distinction is important because the study separates external financial conditions from domestic market movements.

TABLE I
MAIN VARIABLES USED IN THE STUDY

Variable	Transformation	Interpretation
fx_return	$100\Delta \ln S_t$	Daily USD/VND return.
abs_return	$ r_t $	Absolute return volatility proxy.
squared_return	r_t^2	Squared return volatility proxy.
rolling_vol_22d	$\sigma_t^{(22)}$	Preliminary rolling volatility proxy.
vix_change	ΔVIX_t	Global risk-aversion shock.
us10y_change	$\Delta US10Y_t$	U.S. interest-rate shock.
dxy_return	$100\Delta \ln DXY_t$	U.S. dollar strength shock.
oil_return	$100\Delta \ln OIL_t$	Global oil-price shock.
vnindex_return	$100\Delta \ln VNI_t$	Domestic equity-market shock.

C. Event Dummy Variables

Event dummy variables are constructed to control for major policy and crisis episodes during the sample period. The Vietnam exchange-rate policy dummy is separated into two devaluation episodes. The first includes the windows from 11 February 2010 to 11 March 2010 and from 17 August 2010 to 20 August 2010. The second includes the window from 11 February 2011 to 18 February 2011. In addition, the COVID-19 dummy equals one from 1 March 2020 to 31 December 2020, while the Federal Reserve tightening dummy equals one from 1 March 2022 to 31 July 2023. The aggregate crisis dummy is defined as the maximum across the event-period dummies.

The general crisis dummy is used in diagnostic and supplementary specifications to control for broad crisis periods. In the custom EGARCH-X specification and HAC regressions, a separate one-day devaluation dummy is also constructed for 11 February 2010. This dummy is fixed ex ante as a known exchange-rate policy event and is not mechanically selected from the largest residual. This distinction matters because the event-window dummies control for broader crisis periods, while the one-day devaluation dummy captures a specific policy adjustment date.

$$D_t^{crisis} = \max(D_t^{2010}, D_t^{2011}, D_t^{COVID}, D_t^{FED}). \tag{5}$$

D. Preliminary Time-Series Diagnostics

Before estimating volatility models, the transformed variables are examined using descriptive statistics, stationarity tests, autocorrelation tests, volatility-clustering tests, conditional heteroskedasticity tests, normality tests, and multicollinearity checks.

Stationarity is tested using the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. The ADF test has the null hypothesis of a unit root, while the KPSS test has the null hypothesis of stationarity. Using both tests provides a stronger basis for determining whether the transformed return and change variables are suitable for time-series modelling.

Serial dependence in USD/VND returns is examined using the autocorrelation function (ACF), partial autocorrelation function (PACF), and Ljung–Box tests. The Ljung–Box statistic is

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}, \tag{6}$$

where T is the sample size, m is the selected lag length, and $\hat{\rho}_k$ is the sample autocorrelation at lag k . Ljung–Box tests are applied to USD/VND returns to detect mean dependence and to absolute and squared returns to detect volatility clustering.

Conditional heteroskedasticity is examined using the ARCH-LM test. The auxiliary regression is

$$\hat{\varepsilon}_t^2 = a_0 + \sum_{i=1}^q a_i \hat{\varepsilon}_{t-i}^2 + u_t, \tag{7}$$

with the test statistic $LM = TR^2$. A significant ARCH-LM statistic indicates that the conditional variance of USD/VND returns is time-varying and justifies the use of GARCH-family models.

The ARCH-LM test is conducted on both constant-mean residuals and residuals from a compact OLS mean-equation diagnostic. The OLS diagnostic regression includes DXY returns, VIX changes, U.S. 10-year Treasury yield changes, VN-Index returns, and the aggregate crisis dummy. Only the aggregate crisis dummy is used in this diagnostic regression in order to avoid perfect multicollinearity between individual event dummies and the combined crisis dummy. This OLS regression is not treated as a main explanatory model. Its purpose is only to check whether ARCH effects remain after controlling for basic shock variables and crisis periods.

The Jarque–Bera test is used to test normality of USD/VND returns. Since financial returns often exhibit skewness, excess kurtosis, and fat-tailed behavior, rejection of normality supports the use of Student- t innovations in the volatility models. Correlation and variance inflation factor (VIF) checks are also conducted to examine whether the shock variables suffer from severe multicollinearity.

TABLE II
PRE-ESTIMATION DIAGNOSTIC TESTS

Test	Purpose
Descriptive statistics	Examine mean, dispersion, skewness, kurtosis, and extreme values.
ADF and KPSS	Test stationarity of transformed variables.
ACF and PACF	Inspect short-run dependence in USD/VND returns.
Ljung–Box on returns	Test serial correlation in the conditional mean.
Ljung–Box on $ r_t $ and r_t^2	Detect volatility clustering.
ARCH-LM	Test conditional heteroskedasticity.
Jarque–Bera	Test non-normality and fat-tailed behavior.
Correlation and VIF	Check multicollinearity among shock variables.

E. Conditional Mean Specification

Because USD/VND returns exhibit short-run serial dependence, the conditional mean is modelled before estimating the conditional variance. The conditional mean follows an ARMA specification:

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t. \quad (8)$$

The ARMA model is estimated using a grid search over $p = 0, \dots, 5$ and $q = 0, \dots, 5$, excluding the $(0, 0)$ case. Each candidate model is estimated as a SARIMAX model with order $(p, 0, q)$, a constant term, stationarity enforcement, and invertibility enforcement. Candidate models are evaluated using log-likelihood, AIC, BIC, HQIC, convergence status, and Ljung–Box residual diagnostics.

The model-selection rule is deliberately conservative. The selected ARMA model must converge and pass Ljung–Box residual tests at lags 10 and 20 using raw residual p-values. Among the converged models that pass these residual diagnostics, the model with the lowest BIC is selected. Adjusted Ljung–Box statistics with degrees-of-freedom correction are also reported, but the main filtering rule is based on the raw Ljung–Box residual diagnostics.

The selected conditional mean model is ARMA(3,2). Its role is not to serve as the final model of exchange-rate volatility. Instead, it is used as a mean-filtering step. The residuals from the selected ARMA model are passed into the volatility models. Therefore, the empirical strategy is a two-step ARMA–GARCH framework rather than a fully joint ARMA–GARCH maximum-likelihood system.

F. GARCH-family Volatility Models

After the ARMA(3,2) mean equation is selected, the ARMA residuals are used as the input series for GARCH-family volatility modelling. A short burn-in equal to $\max(p, q)$ is removed before fitting the volatility models. Since the selected ARMA model is ARMA(3,2), the first three residual observations are excluded from the GARCH-family estimation stage.

The residual innovation is written as

$$\varepsilon_t = z_t \sqrt{h_t}, \quad z_t \sim t_\nu, \quad (9)$$

where h_t is the conditional variance and z_t follows a standardized Student- t distribution with ν degrees of freedom.

Student- t innovations are used because the diagnostic tests show strong non-normality and fat-tailed behavior in USD/VND returns.

Four GARCH-family specifications are estimated with zero conditional mean because the mean dynamics have already been filtered by ARMA(3,2). The compared specifications are GARCH(1,1)- t , EGARCH(1,1)- t , asymmetric EGARCH(1,1)- t , and GJR-GARCH(1,1)- t .

The standard GARCH(1,1) model is

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}. \quad (10)$$

The EGARCH(1,1) model is

$$\ln h_t = \omega + \beta \ln h_{t-1} + \alpha (|z_{t-1}| - E|z_{t-1}|). \quad (11)$$

The asymmetric EGARCH specification additionally includes a signed shock term:

$$\ln h_t = \omega + \beta \ln h_{t-1} + \alpha (|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1}. \quad (12)$$

The GJR-GARCH(1,1) model is

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I_{t-1} \varepsilon_{t-1}^2 + \beta h_{t-1}, \quad (13)$$

where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ and $I_{t-1} = 0$ otherwise.

Model comparison uses log-likelihood, AIC, BIC, convergence status, residual sanity checks, and volatility diagnostics. A model is considered acceptable only if it converges, has finite likelihood and information criteria, produces reasonably scaled standardized residuals, and does not produce excessive extreme standardized residuals. The core volatility diagnostic requires the Ljung–Box tests on squared standardized residuals at lags 10 and 20 and the ARCH-LM test on standardized residuals to be insignificant at the 5 percent level.

The information criteria are

$$AIC = -2\ell + 2k, \quad BIC = -2\ell + k \ln(T), \quad (14)$$

where ℓ is the maximized log-likelihood, k is the number of estimated parameters, and T is the sample size. The best standard GARCH-family model is selected by the lowest BIC among models that pass the sanity and volatility-diagnostic checks. This stage selects ARMA(3,2)-EGARCH(1,1)- t as the preferred standard volatility model.

It is important that this model passes the core volatility diagnostics but does not pass the full residual diagnostics, because the Ljung–Box tests on standardized residuals remain significant. Therefore, the selected EGARCH model should be interpreted as successfully modelling conditional heteroskedasticity, not as fully eliminating every form of residual serial dependence.

G. Custom EGARCH-X Shock Transmission Model

After selecting ARMA(3,2)-EGARCH(1,1)- t as the preferred baseline volatility model, the analysis estimates custom EGARCH-X specifications to examine shock transmission directly within the conditional variance equation. This stage is separate from the construction of the selected conditional volatility series used later in the HAC and weekly SVAR analyses.

The purpose of the custom EGARCH-X model is to test whether event controls and lagged global or domestic shocks improve the conditional variance specification. Since the standard `arch` implementation does not directly estimate this exact EGARCH-X variance specification with the desired shock controls, the EGARCH-X model is estimated manually by maximum likelihood.

The custom EGARCH-X model uses the ARMA(3,2)-filtered residuals as the innovation series. The shock variables are standardized before entering the variance equation. For each shock variable x_t , the standardized signed shock and absolute shock are defined as

$$z_t^x = \frac{x_t - \bar{x}}{s_x}, \quad |z_t^x| = \left| \frac{x_t - \bar{x}}{s_x} \right|. \quad (15)$$

Lagged signed and lagged absolute shock variables are then constructed:

$$z_{t-1}^x, \quad |z_{t-1}^x|. \quad (16)$$

Lagged shocks are used because USD/VND may not fully absorb global market information on the same trading day, especially under a managed exchange-rate environment and because of timing differences across financial markets.

The custom EGARCH-X variance equation is

$$\ln h_t = \omega + \beta \ln h_{t-1} + \alpha (|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1} + X_t' \delta. \quad (17)$$

where X_t contains the selected shock variables and event controls. Depending on the specification, X_t may contain the one-day devaluation dummy, lagged signed standardized shocks, lagged absolute standardized shocks, the aggregate crisis dummy, or combinations of these variables.

The conditional variance recursion starts from the sample variance of the ARMA residuals. To avoid numerical instability, the log variance is clipped within a bounded range during the recursion. The Student- t log-likelihood is maximized with respect to the variance parameters and the degrees-of-freedom parameter. The parameter β is transformed using the hyperbolic tangent function so that it remains within a stationary range, while the Student- t degrees-of-freedom parameter is transformed to remain above two.

The following EGARCH-X candidate specifications are estimated:

- 1) EGARCH without exogenous variance regressors;
- 2) EGARCH-X with the 11 February 2010 devaluation dummy only;

- 3) EGARCH-X with lagged signed shocks and the crisis dummy;
- 4) EGARCH-X with lagged absolute shocks and the crisis dummy;
- 5) EGARCH-X with lagged signed shocks, the crisis dummy, and the devaluation dummy;
- 6) EGARCH-X with lagged absolute shocks, the crisis dummy, and the devaluation dummy.

The shock vector includes VIX changes, U.S. 10-year Treasury yield changes, DXY returns, oil returns, and VN-Index returns. The first four variables represent global financial shocks, while VN-Index returns represent domestic financial-market conditions.

The custom EGARCH-X models are evaluated using log-likelihood, AIC, BIC, optimizer success, post-burn-in standardized residual behavior, Ljung-Box tests on squared standardized residuals, and ARCH-LM tests. A 60-observation diagnostic burn-in is used when evaluating standardized residuals. The best EGARCH-X specification is selected by the lowest BIC among models passing the sanity and volatility-diagnostic checks.

This selection rule chooses the EGARCH-X model with the 11 February 2010 devaluation dummy only. However, if a selected coefficient is located at an optimizer boundary, its exact magnitude and statistical significance are not interpreted as a precise marginal effect. In that case, the event dummy is interpreted as an event-control indicator showing that the policy adjustment date improves the variance specification, not as a smoothly estimated continuous shock coefficient.

H. Selected Conditional Volatility Series

The selected conditional volatility series used in the supplementary HAC and weekly SVAR analyses is extracted from the ARMA(3,2)-EGARCH(1,1)- t baseline volatility model. It is not extracted from the custom EGARCH-X model. This distinction is important because the baseline EGARCH model and the custom EGARCH-X models serve different purposes. The baseline EGARCH model provides the model-implied volatility series, while the custom EGARCH-X models test whether shock variables enter the conditional variance equation directly.

The fitted conditional variance from the ARMA(3,2)-EGARCH(1,1)- t model is denoted by

$$\hat{h}_t^{sel}. \quad (18)$$

The selected conditional volatility is

$$\hat{\sigma}_t^{sel} = \sqrt{\hat{h}_t^{sel}}. \quad (19)$$

The logarithm of the selected conditional volatility is then constructed as

$$\ell_t^\sigma = \ln(\hat{\sigma}_t^{sel}). \quad (20)$$

The logarithmic transformation is used because it stabilizes the scale of the volatility series and reduces the influence of extreme volatility observations. The resulting variable, ℓ_t^σ , is

used as the dependent variable in the HAC shock-volatility regressions and as the selected-volatility target variable in the weekly SVAR system.

Therefore, the empirical framework separates two roles. The ARMA(3,2)-EGARCH(1,1)- t model is the source of the selected conditional volatility series. The custom EGARCH-X models are estimated separately to evaluate shock transmission in the variance equation.

I. Supplementary HAC Shock-Volatility Regression

As a supplementary reduced-form check, the study estimates HAC shock-volatility regressions. The dependent variable is the logarithm of selected conditional volatility from the ARMA(3,2)-EGARCH(1,1)- t baseline model. These regressions do not replace the EGARCH-X variance-equation analysis and are not interpreted as structural causal estimates. Their purpose is to examine whether lagged shock variables are associated with the model-implied USD/VND volatility series under heteroskedasticity- and autocorrelation-consistent standard errors.

Two HAC specifications are estimated. The signed-shock model is

$$\begin{aligned} \ell_t^\sigma = & c + \beta_1 z_{t-1}^{VIX} + \beta_2 z_{t-1}^{US10Y} + \beta_3 z_{t-1}^{DXY} \\ & + \beta_4 z_{t-1}^{OIL} + \beta_5 z_{t-1}^{VNI} + \eta_1 D_t^{crisis} \\ & + \eta_2 D_t^{dev} + u_t. \end{aligned} \quad (21)$$

The absolute-shock model is

$$\begin{aligned} \ell_t^\sigma = & c + \beta_1 |z_{t-1}^{VIX}| + \beta_2 |z_{t-1}^{US10Y}| + \beta_3 |z_{t-1}^{DXY}| \\ & + \beta_4 |z_{t-1}^{OIL}| + \beta_5 |z_{t-1}^{VNI}| + \eta_1 D_t^{crisis} \\ & + \eta_2 D_t^{dev} + u_t. \end{aligned} \quad (22)$$

The models are estimated by OLS, but inference uses HAC standard errors with five lags. BIC and AIC are reported for model comparison, but the HAC section is interpreted only as supplementary association evidence. A stronger result in the absolute-shock specification would indicate that the magnitude of financial shocks explains USD/VND volatility better than the signed direction of the shocks.

J. Weekly VAR, Granger Causality, and SVAR Analysis

The final empirical stage uses weekly VAR and SVAR models as supplementary dynamic evidence. The purpose is to examine whether global and domestic shocks help predict USD/VND returns or selected conditional volatility, and to trace the dynamic response of the exchange-rate system to structural shocks.

Daily variables are aggregated to weekly frequency using Friday-ending weeks. Weekly shock variables and USD/VND returns are computed by summing daily values within each week. The selected log conditional volatility from the ARMA(3,2)-EGARCH(1,1)- t baseline model is aggregated using the weekly mean:

$$\begin{aligned} x_t^W &= \sum_{d \in W_t} x_d, \\ r_t^W &= \sum_{d \in W_t} r_d, \\ v_t^W &= \frac{1}{n_t} \sum_{d \in W_t} \ell_d^\sigma. \end{aligned} \quad (23)$$

After aggregation, all weekly variables are standardized before VAR/SVAR estimation. The weekly SVAR sample contains 831 weekly observations.

Two systems are estimated. The first system studies shock transmission to weekly USD/VND returns:

$$Y_t^{FX} = \begin{bmatrix} \Delta VIX_t \\ \Delta US10Y_t \\ r_t^{DXY} \\ r_t^{OIL} \\ r_t^{VNI} \\ r_t^{FX} \end{bmatrix}. \quad (24)$$

The second system studies shock transmission to selected conditional volatility:

$$Y_t^{VOL} = \begin{bmatrix} \Delta VIX_t \\ \Delta US10Y_t \\ r_t^{DXY} \\ r_t^{OIL} \\ r_t^{VNI} \\ v_t^W \end{bmatrix}. \quad (25)$$

The reduced-form VAR model is

$$Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + u_t. \quad (26)$$

The lag length is selected using BIC, with AIC used as a fallback if necessary, and with a minimum lag of one. In the implementation, the return system selects four lags, while the selected-volatility system selects one lag.

Before interpreting the SVAR impulse responses, Granger causality tests are applied to examine whether each shock variable contains predictive information for the target variable. These tests are interpreted as predictive evidence only. They do not prove structural causality.

The structural model is specified as

$$Au_t = \varepsilon_t, \quad (27)$$

where u_t is the vector of reduced-form VAR residuals and ε_t is the vector of structural shocks. A recursive short-run (A)-model identification is imposed. The ordering is

$$\begin{aligned} VIX &\rightarrow US10Y \rightarrow DXY \\ &\rightarrow OIL \rightarrow VNINDEX \rightarrow FX/VOL. \end{aligned} \quad (28)$$

This ordering assumes that global financial variables are contemporaneously more exogenous than Vietnam-specific

financial variables and exchange-rate outcomes. VN-Index is placed before the USD/VND return or volatility target because domestic equity-market movements may affect exchange-rate conditions within the week, while USD/VND movements are not assumed to contemporaneously affect global financial variables.

The recursive (A) matrix is lower triangular, with ones on the diagonal, freely estimated elements below the diagonal, and zeros above the diagonal:

$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a_{21} & 1 & 0 & \cdots & 0 \\ a_{31} & a_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ a_{k1} & a_{k2} & a_{k3} & \cdots & 1 \end{bmatrix}. \quad (29)$$

Impulse response functions are computed over a 12-week horizon. Because the SVAR depends on weekly aggregation and recursive ordering assumptions, the SVAR results are treated as supplementary dynamic evidence. The main volatility conclusions are drawn from the ARMA, GARCH-family, EGARCH-X, and HAC results.

K. Hypotheses

Based on the empirical framework, the study tests the following hypotheses.

TABLE III
RESEARCH HYPOTHESES

Hyp.	Statement	Main Test
H_1	USD/VND returns exhibit volatility clustering and conditional heteroskedasticity.	Ljung–Box; ARCH-LM
H_2	USD/VND exchange rate volatility is persistent over time.	GARCH-family and EGARCH-X models
H_3	USD/VND return innovations are fat-tailed and better modelled using Student- t innovations.	Jarque–Bera; Student- t parameter
H_4	Major exchange-rate policy and crisis episodes are associated with shifts in USD/VND volatility.	Event dummies in EGARCH-X and HAC
H_5	Global financial shocks are associated with USD/VND exchange rate volatility.	EGARCH-X; HAC; weekly SVAR
H_6	Absolute shock magnitudes explain USD/VND volatility more strongly than signed shock directions.	Signed vs. absolute-shock models
H_7	Domestic equity-market shocks contribute to USD/VND exchange rate volatility.	VN-Index coefficients

The hypotheses are evaluated sequentially. H_1 to H_3 are tested using diagnostics and GARCH-family models. H_4 to H_6 are tested mainly through EGARCH-X and HAC specifications. H_7 is evaluated through the VN-Index shock variable across the EGARCH-X, HAC, and weekly SVAR sections.

IV. RESULTS

A. Overview of Model Selection Results

This section reports the empirical results from the full modelling pipeline. The analysis proceeds in five stages. First, ARMA models are compared to select the conditional mean specification for USD/VND returns. Second, GARCH-family models are estimated on the ARMA-filtered residuals to

identify the most suitable baseline volatility model. Third, custom EGARCH-X specifications are estimated to examine whether policy events and global or domestic shocks enter the conditional variance equation. Fourth, HAC shock-volatility regressions are used as supplementary reduced-form evidence. Finally, weekly VAR/SVAR models are estimated to examine dynamic shock transmission.

The final model-selection summary is as follows. The selected conditional mean model is ARMA(3,2). The selected baseline GARCH-family volatility model is ARMA(3,2)-EGARCH(1,1)-(t). The selected custom EGARCH-X variance specification is the EGARCH-X model with the 11 February 2010 devaluation dummy only. The selected supplementary HAC model is the absolute-shock specification with one-period lagged standardized absolute shocks. The weekly SVAR models use recursive short-run (A)-model identification, with lag length selected by BIC and a minimum lag of one.

TABLE IV
FINAL MODEL SELECTION SUMMARY

Block	Selected model	Selection rule
ARMA mean baseline	ARMA(3,2)	Lowest BIC among converged ARMA models passing residual diagnostics.
GARCH-family volatility	ARMA(3,2)-EGARCH(1,1)-(t)	Lowest BIC among converged models passing residual sanity and volatility diagnostics.
Custom EGARCH-X variance equation	EGARCH-X with devaluation dummy only	Lowest BIC among custom EGARCH/EGARCH-X models passing sanity checks.
Supplementary HAC regression	HAC absolute shocks, lag 1	Lowest BIC among supplementary HAC specifications.
Supplementary weekly SVAR	Recursive short-run (A)-model SVAR	Lag selected by BIC, with AIC fallback and minimum lag one.

The results should therefore be interpreted carefully. The strongest evidence concerns volatility clustering, fat-tailed innovations, and persistent conditional volatility. The evidence for global and domestic shock transmission is present but selective. It is stronger in the absolute-shock HAC regression and in some richer EGARCH-X specifications than in the BIC-selected EGARCH-X model.

B. Exploratory Data Analysis and Feature Engineering Results

Before conducting formal time-series diagnostics and model estimation, the dataset is examined through exploratory data analysis and feature construction. This step verifies that the cleaned data, transformed variables, event controls, and model-ready sample are consistent with the empirical objective of studying USD/VND exchange rate volatility and shock transmission.

1) *Cleaned Sample and Main Level Variables:* After data cleaning and feature construction, the final model-ready daily sample covers the period from 26 January 2010 to 31 December 2025. The final dataset contains 4,006 daily observations.

This sample is used as the common basis for the preliminary diagnostics, ARMA mean filtering, GARCH-family volatility models, custom EGARCH-X specifications, HAC shock-volatility regressions, and weekly SVAR analysis.

The main level variables include the USD/VND exchange rate, VIX, U.S. 10-year Treasury yield, DXY, VN-Index, and WTI crude oil prices. The USD/VND exchange rate increases from 18,469.00 at the beginning of the final sample to 26,301.95 at the end of the sample. The average USD/VND level is 22,554.19, with a standard deviation of 1,688.21. This pattern indicates a long-run depreciation trend of the Vietnamese dong against the U.S. dollar, but the daily movement is not smooth. Instead, the series contains periods of gradual adjustment interrupted by discrete exchange-rate movements.

TABLE V
CLEANED SAMPLE AND MAIN LEVEL VARIABLES

Variable	Obs.	Mean	Std.	Min.	Max.
usd_vnd	4006	22554.19	1688.21	18469.00	26434.05
vix	4006	18.41	6.89	9.14	82.69
us_10y	4006	2.61	1.00	0.52	4.98
dxy	4006	92.80	9.34	72.93	114.11
vnindex	4006	854.53	350.11	336.73	1784.49
wti_oil	4006	71.58	20.93	-36.98	123.64

The level-variable summary shows that the dataset contains substantial variation in both global and domestic financial conditions. VIX reaches a maximum of 82.69, reflecting the presence of extreme global risk-aversion episodes. The U.S. 10-year Treasury yield ranges from 0.52 to 4.98 percent, capturing both the low-rate period after the global financial crisis and the later monetary-tightening period. DXY also varies considerably, from 72.93 to 114.11, indicating meaningful changes in broad U.S. dollar strength. The VN-Index rises from the lower range of the sample to a maximum of 1,784.49, while WTI crude oil prices contain an extreme negative observation, consistent with the abnormal oil-market episode in 2020.

These level-variable patterns confirm that the sample contains several periods of financial stress and regime-specific movements. Therefore, the data are suitable for studying whether external financial shocks and domestic market movements are associated with USD/VND volatility.

2) *Exchange Rate Returns and Volatility Patterns:* The USD/VND level series is transformed into daily log returns before modelling. This transformation removes the long-run trending behavior of the exchange-rate level and focuses the analysis on short-run exchange-rate movements. The return series shows a low average daily movement but contains several large discrete changes. This is consistent with a managed exchange-rate environment, where daily movements are usually small but policy-related adjustments can generate large return observations.

The largest absolute USD/VND return in the final sample occurs on 14 February 2011, with a return of 6.2139 percent. Several other large return observations occur during the 2010 exchange-rate adjustment period. These large observations are

important because they create fat-tailed behavior and motivate the use of Student- t innovations in the subsequent volatility models.

TABLE VI
LARGEST ABSOLUTE USD/VND RETURN OBSERVATIONS

Date	USD/VND	fx_return	rolling_vol_22d	Event window
2011-02-14	20910.00	6.2139	1.3265	2011 devaluation
2010-02-18	18550.00	-2.3942	0.7042	2010 devaluation
2010-02-24	19075.05	2.3608	0.8904	2010 devaluation
2010-03-09	18660.00	-2.2781	1.4148	2010 devaluation
2010-02-25	18650.00	-2.2535	1.0272	2010 devaluation
2010-02-26	19075.05	2.2535	1.1298	2010 devaluation
2010-03-04	18680.00	-2.1977	1.2368	2010 devaluation
2010-03-11	19095.08	2.1977	1.4869	2010 devaluation

The concentration of extreme returns around exchange-rate adjustment episodes indicates that USD/VND volatility is not purely continuous. It is partly shaped by discrete event periods. This justifies the inclusion of event dummy variables in the EGARCH-X and HAC specifications.

The 22-day rolling volatility series also confirms the presence of time-varying exchange-rate risk. The average rolling volatility is 0.1447, while the maximum reaches 1.4870 during March 2010. The highest rolling-volatility observations are concentrated around the 2010 exchange-rate adjustment window. This provides descriptive evidence of volatility clustering before formal ARCH and GARCH tests are conducted.

TABLE VII
ROLLING VOLATILITY SUMMARY

Variable	Obs.	Mean	Std.	Min.	Max.
rolling_vol_22d	4006	0.1447	0.1567	0.0076	1.4870

These EDA results support the use of conditional volatility models. The USD/VND return series has long quiet periods, extreme observations, and clustered volatility episodes. A constant-variance model would fail to capture these patterns.

3) *Feature Engineering for Return and Shock Variables:* The feature-engineering step converts raw level variables into stationary return or change variables. USD/VND, DXY, WTI oil prices, and VN-Index are transformed into percentage log returns. VIX and the U.S. 10-year Treasury yield are transformed into first differences because they are already index or rate variables. These transformations are used to measure daily shocks rather than long-run levels.

TABLE VIII
FEATURE ENGINEERING TRANSFORMATIONS

Constructed variable	Transformation	Purpose
fx_return	$100\Delta \ln(S_t)$	Measures daily USD/VND movement.
dxy_return	$100\Delta \ln(DXY_t)$	Measures U.S. dollar strength shock.
oil_return	$100\Delta \ln(OIL_t)$	Measures global oil-price shock.
vnindex_return	$100\Delta \ln(VNINDEX_t)$	Measures domestic equity-market shock.
vix_change	ΔVIX_t	Measures global risk-aversion shock.
us10y_change	$\Delta US10Y_t$	Measures U.S. interest-rate shock.
abs_return	$ r_t $	Preliminary volatility proxy.
squared_return	r_t^2	Preliminary volatility proxy.
rolling_vol_22d	$sd_{22}(r_t)$	Preliminary rolling-volatility measure.

The transformed variables are used in different parts of the empirical framework. `fx_return` is the main exchange-rate return series used in the ARMA and GARCH-family models. The global shock variables are `vix_change`, `us10y_change`, `dxy_return`, and `oil_return`. The domestic financial-market shock variable is `vnindex_return`. The preliminary volatility proxies, including `abs_return`, `squared_return`, and `rolling_vol_22d`, are used only for descriptive analysis and volatility-clustering diagnostics.

The final model-ready dataset contains no missing values for the main non-oil transformed variables. Oil-related variables have two missing observations after transformation. These missing observations explain why some model specifications using oil returns have slightly fewer valid observations than the full 4,006-observation sample.

4) *Event Dummy Construction*: Event dummy variables are constructed to control for major exchange-rate policy and crisis periods. The 2010 and 2011 Vietnam exchange-rate devaluation dummies identify periods of discrete USD/VND adjustment. The COVID-19 dummy captures the 2020 pandemic shock period. The Federal Reserve tightening dummy captures the 2022–2023 global monetary-tightening period. The aggregate crisis dummy equals one if any of these event-period dummies equals one.

TABLE IX
EVENT DUMMY CONSTRUCTION RESULTS

Dummy variable	Number of observations
<code>vietnam_fx_devaluation_2010</code>	24
<code>vietnam_fx_devaluation_2011</code>	6
<code>covid_2020</code>	213
<code>fed_hiking_2022_2023</code>	356
<code>crisis_dummy</code>	599

The dummy construction confirms that the final sample contains several important event periods. The aggregate crisis dummy covers 599 daily observations. These event controls are important because large discrete policy or crisis episodes may distort the estimated volatility process if left uncontrolled.

In the custom EGARCH-X and HAC sections, the 11 February 2010 devaluation event is also treated as a specific event-control dummy. This one-day dummy is used because the exploratory analysis shows that exchange-rate adjustment episodes generate some of the largest USD/VND return observations in the sample. It is therefore appropriate to test whether this event improves the conditional variance specification.

5) *Shock Standardization and Lag Construction*: For the EGARCH-X and HAC shock-transmission specifications, shock variables are standardized before being included in the models. Standardization is necessary because the shock variables are measured in different units. For example, VIX changes, U.S. Treasury yield changes, DXY returns, oil returns, and VN-Index returns do not have directly comparable scales.

For each shock variable x_t , the standardized signed shock is defined as

$$Z_t^x = \frac{x_t - \bar{x}}{s_x}, \quad (30)$$

where $\bar{x}$ is the sample mean and s_x is the sample standard deviation. The absolute shock magnitude is then constructed as

$$|Z_t^x| = \left| \frac{x_t - \bar{x}}{s_x} \right|. \quad (31)$$

Lagged versions of the signed and absolute standardized shocks are used in the EGARCH-X and HAC specifications:

$$Z_{t-1}^x, \quad |Z_{t-1}^x|. \quad (32)$$

Lagging the shock variables is appropriate because the USD/VND market may not fully absorb global financial information on the same trading day. This is especially relevant in a managed exchange-rate environment and in a dataset combining Vietnamese and global financial-market variables.

The distinction between signed and absolute shocks is central to the empirical design. Signed shocks test whether the direction of the shock matters. Absolute shocks test whether the magnitude of the shock matters regardless of direction. This allows the study to evaluate whether USD/VND volatility responds more strongly to large disturbances than to positive or negative movements specifically.

6) *Selected Conditional Volatility Feature*: After ARMA mean filtering and GARCH-family model comparison, the ARMA(3,2)-EGARCH(1,1)- t model is selected as the baseline volatility model. The fitted conditional volatility from this model is extracted as the selected conditional volatility series:

$$\hat{\sigma}_t^{sel}. \quad (33)$$

The selected conditional variance and logarithmic transformations are also constructed:

$$\begin{aligned} \hat{h}_t^{sel} &= (\hat{\sigma}_t^{sel})^2, \\ \ell_t^\sigma &= \ln(\hat{\sigma}_t^{sel}), \\ \ell_t^h &= \ln(\hat{h}_t^{sel}). \end{aligned} \quad (34)$$

In the implementation, $\hat{\sigma}_t^{sel}$, $\hat{h}_t^{sel}$, ℓ_t^σ , and ℓ_t^h correspond to `selected_cond_vol`, `selected_cond_var`, `log_selected_cond_vol`, and `log_selected_cond_var`, respectively.

The selected conditional volatility feature contains 4,003 non-missing observations. The first three observations are missing because of the ARMA(3,2) filtering and volatility-estimation alignment. The selected volatility source is fixed as ARMA(3,2)-EGARCH(1,1)- t .

TABLE X
SELECTED CONDITIONAL VOLATILITY SUMMARY

Variable	Obs.	Mean	Std.	Median	Max.
<code>selected_cond_vol</code>	4003	0.1761	0.1988	0.1406	4.2026
<code>selected_cond_var</code>	4003	0.0705	0.5194	0.0198	17.6618
<code>log_selected_cond_vol</code>	4003	-1.9925	0.6728	-1.9620	1.4357
<code>log_selected_cond_var</code>	4003	-3.9851	1.3455	-3.9240	2.8714

The selected conditional volatility series is used as a model-implied measure of USD/VND exchange-rate risk. Its logarithmic form is used in the HAC and weekly SVAR sections to reduce scale instability and limit the influence of extreme volatility observations. This feature-engineering step links the GARCH-family volatility results to the later reduced-form and dynamic shock-transmission analyses.

7) *Weekly Aggregation for SVAR*: The final feature-engineering step aggregates daily variables to weekly frequency for VAR/SVAR analysis. Weekly aggregation reduces high-frequency noise and improves the stability of the multivariate system. This is especially useful because daily USD/VND movements may be affected by market frictions, non-trading days, and policy smoothing.

Daily return and shock variables are aggregated by weekly summation. The selected log conditional volatility is aggregated using the weekly mean. The weekly system contains 831 observations. This weekly dataset is then used to estimate two SVAR systems: one with `fx_return` as the target variable and one with `log_selected_cond_vol` as the target variable.

TABLE XI
FINAL FEATURE SETS USED IN THE EMPIRICAL MODELS

Model block	Main features used
ARMA mean model	<code>fx_return</code> .
GARCH-family models	ARMA(3,2)-filtered residuals of <code>fx_return</code> .
Custom EGARCH-X	ARMA residuals, lagged signed shocks, lagged absolute shocks, crisis dummy, and devaluation dummy.
HAC regression	<code>log_selected_cond_vol</code> , lagged signed shocks, lagged absolute shocks, crisis dummy, and devaluation dummy.
Weekly SVAR	Weekly <code>vix_change</code> , <code>us10y_change</code> , <code>dxy_return</code> , <code>oil_return</code> , <code>vnindex_return</code> , and either <code>fx_return</code> or <code>log_selected_cond_vol</code> .

Overall, the EDA and feature-engineering results show that the dataset is suitable for the empirical strategy. The cleaned sample contains sufficient daily observations, the USD/VND return series displays discrete jumps and time-varying volatility,

the shock variables capture meaningful global and domestic financial variation, and the constructed features support the sequential modelling framework from ARMA mean filtering to GARCH-family volatility modelling, EGARCH-X shock transmission, HAC robustness checks, and weekly SVAR dynamic analysis.

C. Preliminary Time-Series Diagnostic Results

The preliminary diagnostics examine whether the transformed variables satisfy the empirical conditions needed for ARMA, GARCH-family, EGARCH-X, HAC, and SVAR modelling. The tests focus on stationarity, short-run mean dependence, volatility clustering, conditional heteroskedasticity, non-normality, and multicollinearity among the shock variables.

1) *Descriptive Statistics*: Table XII reports descriptive statistics for the main transformed variables. The final model-ready sample contains 4,006 daily observations from 26 January 2010 to 31 December 2025. Oil returns contain 4,004 valid observations because of missing values after transformation.

TABLE XII
DESCRIPTIVE STATISTICS FOR MODEL VARIABLES

Variable	Obs.	Mean	Std.	Skew.	Kurt.
<code>fx_return</code>	4006	0.0088	0.2129	6.4640	213.1828
<code>dxy_return</code>	4006	0.0057	0.4474	-0.0348	4.5347
<code>vix_change</code>	4006	-0.0026	1.8739	1.9111	32.2619
<code>us10y_change</code>	4006	0.0001	0.0535	0.0043	4.6246
<code>vnindex_return</code>	4006	0.0327	1.1763	-0.8105	7.2700
<code>oil_return</code>	4004	0.0113	2.7136	1.0326	43.5383

The average USD/VND return is close to zero, at 0.0088 percent, which is consistent with daily exchange rate returns. However, the distribution of USD/VND returns is highly non-normal. The series has positive skewness of 6.4640 and Pearson kurtosis of 213.1828, indicating a heavy right tail and a high frequency of extreme observations. This pattern is consistent with a managed exchange-rate environment, where daily movements are often small but discrete adjustment episodes can generate large tail observations.

The standard deviation of `fx_return` is 0.2129, which is lower than the standard deviations of `vnindex_return` and `oil_return`. This reflects the relatively managed nature of the USD/VND exchange rate compared with equity and commodity prices. Nevertheless, the extreme kurtosis of USD/VND returns shows that exchange rate adjustments are not normally distributed. The high kurtosis of VIX changes and oil returns also confirms that the external shock variables contain large tail events. These distributional properties support the use of fat-tailed innovations in the volatility models.

Table XIII reports the event dummy counts. The aggregate crisis dummy equals one for 599 observations, covering the Vietnam exchange-rate devaluation windows, the COVID-19 period, and the 2022–2023 Federal Reserve tightening period.

TABLE XIII
EVENT DUMMY COUNTS

Dummy variable	Number of days
vietnam_fx_devaluation_2010	24
vietnam_fx_devaluation_2011	6
covid_2020	213
fed_hiking_2022_2023	356
crisis_dummy	599

2) *Stationarity Tests*: Table XIV reports the ADF and KPSS stationarity tests. The ADF test rejects the unit-root null for all transformed variables at the 1 percent level. The KPSS test does not reject the stationarity null for any variable, with reported p-values at the upper bound of 0.1000. The two tests therefore provide consistent evidence that the transformed return and change variables are stationary.

TABLE XIV
ADF AND KPSS STATIONARITY TEST RESULTS

Variable	ADF	ADF p	Lag	KPSS	KPSS p	Lag	Result
fx_return	-32.4961	0.0000	3	0.1303	0.1000	5	I(0)
dxy_return	-62.4604	0.0000	0	0.0797	0.1000	11	I(0)
vix_change	-15.1685	0.0000	25	0.0109	0.1000	34	I(0)
us10y_change	-47.1905	0.0000	1	0.2569	0.1000	8	I(0)
vnindex_return	-41.5835	0.0000	1	0.0724	0.1000	8	I(0)
oil_return	-11.9710	0.0000	23	0.0692	0.1000	1	I(0)

These results indicate that the transformed variables can be used directly in the ARMA, GARCH-family, EGARCH-X, HAC, and VAR/SVAR models without additional differencing. This is important because applying volatility models to non-stationary level variables could produce misleading persistence and variance estimates.

3) *Autocorrelation and Mean Dependence*: The ACF and PACF diagnostics indicate short-run dependence in USD/VND returns. Using the 95 percent confidence bound of approximately ± 0.0310 , significant ACF lags appear at lags 1, 3, 4, 5, 8, 9, 10, 12, 13, and 18. Significant PACF lags appear at lags 1, 2, 3, 4, 5, 8, 10, and 12. These patterns suggest that the USD/VND return series is not pure white noise in the conditional mean.

Table XV confirms this result. The Ljung–Box test rejects the null hypothesis of no autocorrelation at lags 5, 10, and 20, with p-values equal to 0.0000.

TABLE XV
LJUNG–BOX TEST ON USD/VND RETURNS

Lag	Q-stat.	p-value	Result
5	108.8015	0.0000	Autocorrelation detected
10	146.8841	0.0000	Autocorrelation detected
20	181.1506	0.0000	Autocorrelation detected

The presence of serial correlation means that the conditional mean cannot be ignored. Therefore, the modelling strategy first filters short-run mean dynamics using an ARMA specification before estimating the conditional variance process.

4) *Volatility Clustering and ARCH Effects*: Table XVI reports Ljung–Box tests on absolute and squared USD/VND returns. The tests strongly reject the null hypothesis of no serial correlation for both volatility proxies at all reported lags.

TABLE XVI
LJUNG–BOX TESTS ON ABSOLUTE AND SQUARED RETURNS

Variable	Property	Lag	Q-stat.	Conclusion
abs_return	$ r_t $	5	1007.3634	Clustering detected
abs_return	$ r_t $	10	1482.5224	Clustering detected
abs_return	$ r_t $	20	1685.4318	Clustering detected
squared_return	r_t^2	5	48.8911	Clustering detected
squared_return	r_t^2	10	83.5073	Clustering detected
squared_return	r_t^2	20	90.3285	Clustering detected

Note: All reported p-values are 0.0000.

These results show that the magnitude of exchange-rate movements is persistent over time. Large USD/VND movements tend to be followed by further large movements, while tranquil periods tend to persist. This is the standard empirical motivation for using GARCH-family models instead of constant-variance models.

The ARCH-LM results in Table XVII provide further evidence of conditional heteroskedasticity. The test rejects the null hypothesis of no ARCH effects for both constant-mean residuals and residuals from the compact OLS mean-equation diagnostic.

TABLE XVII
ARCH-LM TEST RESULTS

Residual source	Lags	ARCH-LM stat.	ARCH-LM p-value	F-stat.	Conclusion
Constant mean residuals	10	59.5223	0.0000	6.0256	ARCH effect detected
OLS mean equation residuals	10	61.3004	0.0000	6.2084	ARCH effect detected

Note: The F-test p-values are also 0.0000 for both residual sources.

The OLS mean-equation diagnostic includes dxy_return, vix_change, us10y_change, vnindex_return, and crisis_dummy. The design matrix has full rank and a condition number of 36.7073, indicating no singular-matrix problem in this aggregate-dummy specification. However, the model has very low explanatory power, with $R^2 = 0.0052$ and adjusted $R^2 = 0.0040$. This confirms that the OLS mean equation is not a sufficient model for USD/VND volatility. Its role is only diagnostic.

Table XVIII reports Ljung–Box tests on squared OLS residuals. The null hypothesis of no serial correlation in squared residuals is rejected at lags 5, 10, and 20. This confirms that volatility clustering remains after the compact mean equation.

TABLE XVIII
LJUNG–BOX TESTS ON SQUARED OLS RESIDUALS

Lag	Q-stat.	p-value	Conclusion
5	51.0218	0.0000	ARCH/volatility clustering remains
10	86.9527	0.0000	ARCH/volatility clustering remains
20	94.2665	0.0000	ARCH/volatility clustering remains

Taken together, the Ljung–Box tests on volatility proxies, the ARCH-LM tests, and the squared-residual diagnostics provide

strong evidence that USD/VND volatility is time-varying. Therefore, GARCH-family volatility models are required.

5) *Normality Test*: Table XIX reports the Jarque–Bera normality test for USD/VND returns. The null hypothesis of normality is strongly rejected, with a Jarque–Bera statistic of 7,383,227.0371 and a p-value of 0.0000.

TABLE XIX
JARQUE–BERA NORMALITY TEST FOR USD/VND RETURNS

Variable	JB stat.	p-value	Skew.	Kurt.	Conclusion
fx_return	7383227.0371	0.0000	6.4616	212.9190	Non-normal

The extremely high kurtosis indicates that large exchange-rate movements occur far more frequently than under a Gaussian distribution. Therefore, Student-*t* innovations are more appropriate than normal innovations in the GARCH-family models.

6) *Correlation and Multicollinearity Diagnostics*: Table XX reports the pairwise correlations among the main return and shock variables. The correlations among the explanatory shock variables are generally moderate. The strongest correlations are between *vix_change* and *us10y_change* at -0.2336 , between *vix_change* and *oil_return* at -0.2176 , and between *us10y_change* and *oil_return* at 0.1705 . These values do not indicate severe pairwise multicollinearity.

TABLE XX
CORRELATION MATRIX OF MAIN MODEL VARIABLES

	fx	dxy	vix	us10y	vni	oil
fx_return	1.0000	0.0093	0.0026	0.0034	-0.0635	-0.0093
dxy_return	0.0093	1.0000	0.0854	0.1561	0.0013	-0.1096
vix_change	0.0026	0.0854	1.0000	-0.2336	-0.0757	-0.2176
us10y_change	0.0034	0.1561	-0.2336	1.0000	0.0423	0.1705
vnindex_return	-0.0635	0.0013	-0.0757	0.0423	1.0000	0.0445
oil_return	-0.0093	-0.1096	-0.2176	0.1705	0.0445	1.0000

The contemporaneous correlation between USD/VND returns and most shock variables is small. This does not imply that the shocks are irrelevant. It only indicates that simple static correlation is insufficient to capture exchange-rate risk transmission. This supports the use of conditional volatility models and dynamic VAR/SVAR analysis.

Table XXI reports VIF values for the main explanatory variables. All VIF values are close to one and far below conventional warning thresholds.

TABLE XXI
VARIANCE INFLATION FACTOR RESULTS

Variable	VIF
dxy_return	1.0575
vix_change	1.1117
us10y_change	1.1196
vnindex_return	1.0083
oil_return	1.0839

The VIF results indicate that the shock variables can be included jointly in the EGARCH-X and HAC specifications without serious multicollinearity concerns.

7) *Summary of Diagnostic Evidence*: The preliminary diagnostics support the modelling strategy. The transformed variables are stationary. USD/VND returns exhibit short-run serial dependence, justifying ARMA mean filtering. Absolute and squared returns show strong volatility clustering. ARCH-LM tests confirm conditional heteroskedasticity. The Jarque–Bera test rejects normality and supports Student-*t* innovations. Correlation and VIF diagnostics show no severe multicollinearity among the shock variables.

Overall, the diagnostic evidence supports an ARMA mean specification combined with GARCH-family volatility models under Student-*t* errors. It also motivates the extension to EGARCH-X shock-transmission models and the supplementary HAC and weekly SVAR analyses.

D. Conditional Mean Model Selection

Before estimating conditional volatility models, alternative ARMA specifications are compared to model short-run dependence in USD/VND returns. This step is necessary because GARCH-type models should be applied to innovations after predictable mean dynamics have been filtered out.

Table XXII reports the main ARMA model-selection results. Among converged ARMA models passing the residual diagnostic rule, ARMA(3,2) provides the lowest AIC, BIC, and HQIC. It achieves an AIC of -1146.615 , a BIC of -1102.546 , and an HQIC of -1130.995 . The Ljung–Box p-values for raw residuals are 0.325 at lag 10 and 0.136 at lag 20, indicating that the model removes most linear serial dependence in the conditional mean.

TABLE XXII
ARMA MEAN EQUATION SELECTION

Model	AIC	BIC	HQIC	LB(10)	LB(20)	ARCH-LM
ARMA(3,2)	-1146.615	-1102.546	-1130.995	0.325	0.136	0.000
ARMA(2,3)	-1144.639	-1100.570	-1129.019	0.266	0.101	0.000
ARMA(2,4)	-1145.384	-1095.019	-1127.532	0.366	0.159	0.000
ARMA(2,5)	-1144.943	-1088.283	-1124.860	0.439	0.178	0.000
ARMA(5,2)	-1144.941	-1088.281	-1124.858	0.349	0.154	0.000

Although the ARMA(3,2) model passes the raw Ljung–Box residual tests, the adjusted Ljung–Box p-values and the ARCH-LM test remain significant. This means that ARMA(3,2) is useful as a conditional mean filter, but it does not remove conditional heteroskedasticity. Therefore, the selected ARMA model is not treated as the final model. It is used as the first step in a two-step ARMA–GARCH framework.

The significant ARCH-LM statistic after ARMA filtering confirms that the remaining variation in USD/VND returns is dominated by conditional variance dynamics rather than additional mean predictability. The next stage therefore estimates GARCH-family models using the ARMA-filtered residuals.

E. GARCH-family Model Comparison

After selecting ARMA(3,2), the residuals from this model are used as the input series for GARCH-family volatility models. The volatility models are estimated with zero conditional mean because the short-run mean dynamics have already been filtered out. The residual sample used for this stage contains 4,003 observations.

Table XXIII compares GARCH(1,1), EGARCH(1,1), asymmetric EGARCH(1,1), and GJR-GARCH(1,1), all estimated under Student-*t* innovations.

TABLE XXIII
GARCH-FAMILY MODEL FIT COMPARISON

ID	Model	LogLik	AIC	BIC
M1	ARMA(3,2)-EGARCH(1,1)-(t)	3346.387	-6684.774	-6659.595
M2	ARMA(3,2)-EGARCH(1,1)-asym-(t)	3346.662	-6683.323	-6651.849
M3	ARMA(3,2)-GJR-GARCH(1,1)-(t)	3146.659	-6283.319	-6251.845
M4	ARMA(3,2)-GARCH(1,1)-(t)	3140.889	-6273.777	-6248.598

The asymmetric EGARCH model has the highest log-likelihood, but it requires an additional parameter and is penalized by AIC and BIC. The standard EGARCH(1,1)-(t) specification has the lowest AIC and BIC. Therefore, ARMA(3,2)-EGARCH(1,1)-(t) is selected as the preferred baseline volatility model.

Table XXIV reports residual diagnostics for the GARCH-family models. The selected EGARCH model passes the core volatility diagnostics: the Ljung–Box p-value on squared standardized residuals is 0.996 at lag 10 and approximately 1.000 at lag 20, while the ARCH-LM p-value is 0.996. This indicates that the EGARCH model successfully absorbs the remaining ARCH effects in the ARMA-filtered residuals.

TABLE XXIV
RESIDUAL DIAGNOSTICS OF GARCH-FAMILY MODELS

ID	LB Res.	LB Sq.	ARCH	Vol. pass	Full pass
M1	0.000	0.996	0.996	Yes	No
M2	0.000	0.996	0.996	Yes	No
M3	0.000	0.000	0.000	No	No
M4	0.000	0.000	0.000	No	No

Note: LB Res. and LB Sq. are Ljung–Box p-values for standardized residuals and squared standardized residuals. ARCH denotes the ARCH-LM p-value.

The GARCH(1,1)-(t) and GJR-GARCH(1,1)-(t) models perform substantially worse. Their log-likelihood values are lower, their AIC and BIC values are less favorable, and they fail the volatility diagnostics. This means that the symmetric GARCH and GJR-GARCH specifications do not adequately capture the volatility clustering in USD/VND returns.

The selected EGARCH model produces reasonably scaled standardized residuals, with mean -0.052 and standard deviation 0.960 . However, the Ljung–Box tests on standardized residuals remain significant at lags 10 and 20. Therefore, the model should not be described as fully eliminating all serial dependence. Its strength is more specific: it captures conditional heteroskedasticity and volatility persistence effectively.

The selected EGARCH volatility series is used as the model-implied conditional volatility measure in the supplementary HAC and weekly SVAR analyses. The extracted series contains 4,003 non-missing conditional-volatility observations. Its mean is 0.1761 , its median is 0.1406 , and its maximum is 4.2026 . The logarithm of selected conditional volatility is used in later regressions to stabilize scale.

F. Custom EGARCH-X Shock Transmission Results

After selecting the standard EGARCH volatility model, the analysis estimates custom EGARCH-X specifications to examine whether event controls and lagged shock variables enter the conditional variance equation. This stage addresses the main shock-transmission question. The custom EGARCH-X models are estimated using ARMA(3,2) residuals as the filtered innovation series.

Six custom EGARCH/EGARCH-X specifications are compared: a no-(X) EGARCH model, a devaluation-dummy-only model, signed-shock models, absolute-shock models, and versions of the signed and absolute shock models that additionally include the 11 February 2010 devaluation dummy.

Table XXV reports the model comparison. All specifications converge and pass the core volatility diagnostics. Based on BIC, the selected EGARCH-X model is the devaluation-dummy-only specification. This model improves the BIC from -6643.507 in the no-(X) EGARCH model to -6650.735 . The model with lagged absolute shocks plus the devaluation dummy has the highest log-likelihood and the lowest AIC, but its BIC is weaker because it uses more parameters.

TABLE XXV
CUSTOM EGARCH-X MODEL COMPARISON

Model	Par.	LogLik	AIC	BIC
No (X)	5	3342.487	-6674.974	-6643.507
Devaluation dummy only	6	3350.247	-6688.495	-6650.735
Absolute shocks	11	3353.075	-6684.150	-6614.927
Absolute shocks + devaluation	12	3360.576	-6697.151	-6621.635
Signed shocks	11	3349.025	-6676.051	-6606.827
Signed shocks + devaluation	12	3355.497	-6686.995	-6611.478

Note: Par. denotes the number of estimated parameters. Bold values indicate the best model according to each criterion.

The BIC result implies that the strongest EGARCH-X evidence is not broad evidence that every global shock improves the variance specification. Instead, the most parsimonious improvement comes from controlling for the February 2010 devaluation episode. The richer absolute-shock model provides useful additional information, but it is not the BIC-selected specification.

Table XXVI reports the key coefficient estimates from the custom EGARCH-X models. In the BIC-selected devaluation-dummy-only model, the absolute shock term is positive and significant, with a coefficient of 0.531 and a p-value below 0.01 . The log-variance persistence parameter is also highly significant, with a coefficient of 0.958 . The Student-*t* degrees-of-freedom parameter is 2.781 , confirming very heavy-tailed innovations.

TABLE XXVI
KEY CUSTOM EGARCH-X COEFFICIENT ESTIMATES

Specification	Parameter	Coef.	SE	t-stat.	p-value
Selected	ω	-0.164	0.027	-6.021	0.000
Selected	α	0.531	0.044	12.118	0.000
Selected	γ	0.017	0.021	0.819	0.413
Selected	β	0.958	0.006	149.880	0.000
Selected	ν	2.781	0.146	19.077	0.000
Selected	Devaluation dummy	3.000	—	—	—
Abs. shocks	$ Z_{t-1}^{OIL} $	0.036	0.017	2.148	0.032
Abs. + dev.	$ Z_{t-1}^{DXY} $	0.052	0.029	1.788	0.074
Abs. + dev.	$ Z_{t-1}^{OIL} $	0.035	0.018	1.926	0.054
Abs. + dev.	Crisis dummy	-0.084	0.024	-3.520	0.000

Note: The devaluation dummy coefficient in the selected EGARCH-X model reaches the optimizer boundary. Its exact coefficient magnitude should not be interpreted as a normal marginal effect.

The asymmetry coefficient γ is positive but statistically insignificant in the selected EGARCH-X model. This indicates that USD/VND volatility is driven more by the magnitude of shocks and volatility persistence than by systematic sign asymmetry. Therefore, the paper should not claim strong evidence of asymmetric volatility transmission.

The devaluation dummy enters the selected model at the optimizer boundary. Its coefficient is fixed at the upper boundary value of 3.000, and standard errors are not available. This means the dummy should be interpreted cautiously. The correct interpretation is that the February 2010 devaluation date represents an exceptional exchange-rate event that improves model fit. It should not be interpreted as a precisely estimated continuous marginal effect.

The richer shock specifications provide selective evidence of global shock transmission. In the absolute-shock model, lagged absolute oil shocks are positive and significant at the 5 percent level. In the absolute-shock-plus-devaluation model, lagged absolute DXY and oil shocks are positive and weakly significant at the 10 percent level. These results suggest that the magnitude of selected global shocks, especially oil shocks and to a weaker extent DXY shocks, contains information about USD/VND volatility. However, this evidence is weaker than the evidence for volatility persistence, fat tails, and the devaluation event.

The signed-shock specifications provide little evidence that the direction of global or domestic shocks directly affects USD/VND volatility. Most signed shock coefficients are statistically insignificant. Therefore, the evidence supports the view that shock magnitudes are more informative than signed directions.

The crisis dummy is negative and significant in the richer specifications. This should not be interpreted as evidence that crisis periods reduce economic risk. Instead, after controlling for lagged shock magnitudes, EGARCH dynamics, and the devaluation event, the crisis dummy captures the remaining average shift in log variance during the defined crisis windows. In a managed exchange-rate environment, policy smoothing can reduce observed daily exchange-rate volatility even when underlying macro-financial pressure is high.

Overall, the custom EGARCH-X results support three conclusions. First, USD/VND volatility is persistent and fat-tailed. Second, the February 2010 devaluation event is an important volatility episode. Third, global shock transmission is selective rather than uniform, with the clearest EGARCH-X shock evidence coming from absolute oil shocks and weaker evidence from absolute DXY shocks.

G. Extreme Residual Diagnostics

The extreme residual table confirms that large standardized residuals are concentrated around known exchange-rate adjustment episodes and other unusual market dates. In the no-(X) EGARCH model, the 11 February 2010 devaluation event produces the largest full-sample standardized residual, with an absolute value above 20. After including the devaluation dummy, this extreme observation is absorbed more effectively by the variance specification, and the full-sample maximum absolute standardized residual falls.

This result supports the inclusion of the devaluation dummy as an event control. However, it also reinforces the need for caution. The devaluation dummy is important because it captures a large discrete policy-related exchange-rate movement, not because it provides a general continuous shock-transmission coefficient.

H. HAC Shock-Volatility Regression Results

After the GARCH-family and EGARCH-X stages, supplementary HAC shock-volatility regressions are estimated. The dependent variable is the logarithm of selected conditional volatility extracted from the ARMA(3,2)-EGARCH(1,1)-(t) model. This point is important: in the current output, the selected conditional volatility series used for HAC and weekly SVAR comes from the baseline EGARCH model, not from the custom EGARCH-X specification.

The HAC regressions are not structural causal models. They are reduced-form checks that test whether lagged shock variables are associated with the model-implied USD/VND volatility series under heteroskedasticity- and autocorrelation-consistent standard errors.

Two specifications are estimated. The first uses lagged signed standardized shocks. The second uses lagged absolute standardized shocks. Table XXVII shows that the absolute-shock model performs better. Its adjusted R^2 is 0.0333, compared with 0.0036 for the signed-shock model. It also has lower AIC and BIC values.

TABLE XXVII
HAC SHOCK-VOLATILITY REGRESSION SUMMARY

Model	Obs.	R^2	Adj. R^2	AIC	BIC
HAC absolute shocks	3997	0.0350	0.0333	7999.469	8049.816
HAC signed shocks	3997	0.0054	0.0036	8120.275	8170.621

Although the explanatory power of the absolute-shock model is still modest, the improvement over the signed-shock model

is meaningful. This supports the interpretation that exchange-rate volatility responds more to the magnitude of financial disturbances than to their signed direction.

Table XXVIII reports the HAC coefficient estimates. In the signed-shock model, none of the lagged signed shock variables is statistically significant. The only strongly significant variable is the 11 February 2010 devaluation dummy.

The absolute-shock model produces stronger evidence. Lagged absolute U.S. 10-year Treasury yield changes, DXY returns, oil returns, and VN-Index returns are all positive and statistically significant at the 1 percent level. This indicates that larger U.S. interest-rate movements, stronger U.S. dollar movements, larger oil-market disturbances, and larger domestic equity-market movements are associated with higher estimated USD/VND conditional volatility.

The absolute VIX coefficient is negative but statistically insignificant. Therefore, the HAC results do not provide robust evidence that VIX changes independently explain USD/VND volatility after controlling for U.S. yields, DXY, oil, VN-Index movements, and event dummies.

The devaluation dummy is positive and strongly significant in both HAC models. This confirms that the 11 February 2010 devaluation episode is associated with a higher level of estimated conditional volatility. Unlike the custom EGARCH-X coefficient, the HAC coefficient is not at an optimizer boundary, so it provides cleaner reduced-form evidence that the event is strongly associated with volatility.

The crisis dummy is negative and weakly significant in the absolute-shock model. As in the EGARCH-X results, this should be interpreted as a conditional control effect, not as evidence that crisis periods are economically stabilizing. After controlling for shock magnitudes and the devaluation event, observed daily exchange-rate volatility may appear lower during some crisis periods because of exchange-rate smoothing and policy intervention.

Overall, the HAC results support the hypothesis that absolute shock magnitudes explain USD/VND volatility better than signed shock directions. The strongest HAC evidence comes from U.S. interest-rate shocks, DXY movements, oil shocks, VN-Index movements, and the devaluation event. However, because the HAC regression uses a generated volatility series as the dependent variable, the results should be treated as supplementary association evidence rather than structural causality.

I. Weekly SVAR and Dynamic Shock Transmission

The final stage uses weekly VAR/SVAR models to examine dynamic shock transmission. The daily data are aggregated to weekly frequency to reduce high-frequency noise and improve system stability. Two systems are estimated: one with weekly USD/VND returns as the target variable and one with weekly average log selected conditional volatility as the target variable.

Table XXIX reports the weekly SVAR setup. The return system contains 831 weekly observations, six variables, and four lags. The volatility system contains 831 weekly observa-

tions, six variables, and one lag. Both systems are successfully estimated.

TABLE XXIX
WEEKLY SVAR MODEL SUMMARY

Target	Obs.	Variables	Lag	Fit status
fx_return	831	6	4	OK
log_selected_cond_vol	831	6	1	OK

Before interpreting impulse responses, Granger causality tests are used to examine whether lagged shock variables contain predictive information for the target variables. These tests are predictive tests, not structural causality tests.

Table XXX reports the results. In the weekly return system, only U.S. 10-year Treasury yield changes significantly predict USD/VND returns at the 5 percent level, with a p-value of 0.000015. VIX, DXY, oil, and VN-Index returns are not significant predictors of weekly USD/VND returns in this system.

In the weekly volatility system, only oil returns significantly predict log selected conditional volatility at the 5 percent level, with a p-value of 0.0266. VIX, U.S. Treasury yield changes, DXY, and VN-Index returns are not significant predictors of selected volatility in the weekly system.

TABLE XXX
WEEKLY SVAR GRANGER CAUSALITY TESTS

Target	Shock	Lag	p-value	Sig. 5%
fx_return	vix_change	4	0.1395	No
fx_return	us10y_change	4	0.0000	Yes
fx_return	dxy_return	4	0.0980	No
fx_return	oil_return	4	0.3165	No
fx_return	vnindex_return	4	0.1860	No
log_selected_cond_vol	vix_change	1	0.8447	No
log_selected_cond_vol	us10y_change	1	0.2456	No
log_selected_cond_vol	dxy_return	1	0.5250	No
log_selected_cond_vol	oil_return	1	0.0266	Yes
log_selected_cond_vol	vnindex_return	1	0.6393	No

Note: Granger causality tests evaluate predictive content within the weekly VAR system. Statistical significance does not prove structural causality.

The structural impulse response results provide additional dynamic evidence. Table XXXI summarizes the external-shock responses at selected horizons. For USD/VND returns, the largest short-run response is to U.S. 10-year Treasury yield shocks, with a response of 0.1545 at horizon one. VIX shocks produce a positive impact response of 0.0859, which remains positive through the first four weeks before fading. DXY shocks peak at horizon one, while oil shocks generate a negative response at horizon one. VN-Index shocks generate a negative impact response.

For selected conditional volatility, the response to VIX shocks is positive and persistent but small, peaking at 0.0346 at horizon one. Oil shocks produce the largest absolute volatility response, reaching -0.0906 at horizon one. DXY, VN-Index, and U.S. yield shocks generate smaller and mostly fading responses.

TABLE XXVIII
HAC SHOCK-VOLATILITY REGRESSION COEFFICIENTS

Model	Variable	Coef.	HAC SE	t-stat.	p-value	Sig.
Signed shocks	Constant	-1.990	0.025	-80.991	0.000	***
Signed shocks	Z_{t-1}^{VIX}	-0.006	0.010	-0.621	0.535	
Signed shocks	Z_{t-1}^{US10Y}	-0.012	0.011	-1.095	0.274	
Signed shocks	Z_{t-1}^{DXY}	-0.016	0.010	-1.547	0.122	
Signed shocks	Z_{t-1}^{OIL}	-0.004	0.009	-0.405	0.686	
Signed shocks	Z_{t-1}^{VNI}	-0.021	0.013	-1.591	0.112	
Signed shocks	Crisis dummy	-0.096	0.087	-1.102	0.270	
Signed shocks	Devaluation dummy	1.247	0.086	14.424	0.000	***
Absolute shocks	Constant	-2.183	0.037	-59.528	0.000	***
Absolute shocks	$ Z_{t-1}^{VIX} $	-0.024	0.021	-1.157	0.247	
Absolute shocks	$ Z_{t-1}^{US10Y} $	0.086	0.022	3.990	0.000	***
Absolute shocks	$ Z_{t-1}^{DXY} $	0.083	0.017	4.956	0.000	***
Absolute shocks	$ Z_{t-1}^{OIL} $	0.064	0.015	4.139	0.000	***
Absolute shocks	$ Z_{t-1}^{VNI} $	0.073	0.021	3.410	0.001	***
Absolute shocks	Crisis dummy	-0.169	0.089	-1.893	0.058	*
Absolute shocks	Devaluation dummy	1.231	0.089	13.851	0.000	***

Note: The dependent variable is $\log(\hat{\sigma}_t^{sel})$, where $\hat{\sigma}_t^{sel}$ is the selected conditional volatility from the ARMA(3,2)-EGARCH(1,1)-(t) model. HAC standard errors are computed using five lags. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

TABLE XXXI
STRUCTURAL IRF SUMMARY AT SELECTED HORIZONS

Target	Shock	H0	H1	H4	H12
fx_return	vix_change	0.0859	0.0438	0.0481	-0.0002
fx_return	us10y_change	0.0611	0.1545	0.0839	0.0017
fx_return	dxy_return	0.0123	0.0801	-0.0017	0.0013
fx_return	oil_return	-0.0069	-0.0694	-0.0050	0.0005
fx_return	vnindex_return	-0.1285	-0.0304	0.0606	0.0002
log_selected_cond_vol	vix_change	0.0173	0.0346	0.0237	0.0058
log_selected_cond_vol	us10y_change	0.0130	-0.0182	-0.0076	-0.0019
log_selected_cond_vol	dxy_return	-0.0298	-0.0335	-0.0169	-0.0042
log_selected_cond_vol	oil_return	-0.0538	-0.0906	-0.0556	-0.0137
log_selected_cond_vol	vnindex_return	-0.0138	-0.0208	-0.0124	-0.0030

Note: Horizons are measured in weeks. The table reports point estimates of structural impulse responses. Confidence bands are not reported, so the IRF results should be interpreted descriptively.

The SVAR results provide selective support for the shock-transmission interpretation. The Granger causality tests show that U.S. interest-rate shocks help predict weekly USD/VND returns, while oil shocks help predict weekly selected conditional volatility. The IRF estimates suggest that VIX and U.S. yield shocks have short-run effects on USD/VND returns, and that VIX and oil shocks are dynamically related to selected volatility. However, because most Granger tests are insignificant and the IRF table does not include confidence intervals, the SVAR evidence should remain supplementary.

J. Hypothesis Evaluation

The empirical results allow the hypotheses to be evaluated as follows.

TABLE XXXII
SUMMARY OF HYPOTHESIS EVALUATION

Hyp.	Evidence	Conclusion
H_1	Ljung–Box tests on absolute and squared returns and ARCH-LM tests are strongly significant.	Supported.
H_2	EGARCH persistence parameter is large and highly significant.	Supported.
H_3	Jarque–Bera rejects normality; Student- t degrees-of-freedom estimates are low and significant.	Supported.
H_4	The devaluation dummy improves the EGARCH-X specification and is significant in HAC regressions, but reaches the optimizer boundary in EGARCH-X.	Partially supported as event-control evidence.
H_5	Absolute oil and DXY shocks matter in richer EGARCH-X models; U.S. yield, DXY, oil, and VN-Index shocks matter in HAC; SVAR evidence is selective.	Partially supported.
H_6	Absolute-shock HAC model strongly outperforms signed-shock HAC model; signed EGARCH-X shock coefficients are mostly insignificant.	Supported.
H_7	VN-Index absolute shocks are significant in HAC but not robust across EGARCH-X and SVAR.	Weakly supported.

The strongest hypotheses are H_1 , H_2 , H_3 , and H_6 . The evidence clearly shows volatility clustering, persistence, fat-tailed innovations, and the stronger role of shock magnitudes relative to signed shock directions. H_4 is supported only as event-control evidence because the devaluation dummy is important but reaches the optimizer boundary in EGARCH-X. H_5 is partially supported because global shock transmission is present but not uniform across all models. H_7 receives weak support because domestic equity-market shocks are significant in the HAC absolute-shock model but not consistently significant in EGARCH-X or SVAR.

K. Overall Results Summary

The full set of results leads to four main conclusions. First, USD/VND returns display the standard empirical features that justify volatility modelling: stationarity after transformation, short-run serial dependence, volatility clustering, conditional

heteroskedasticity, extreme kurtosis, and fat-tailed innovations. Second, the ARMA(3,2)-EGARCH(1,1)-(t) model is the preferred baseline volatility specification. It provides the best BIC among the standard GARCH-family models and successfully removes remaining ARCH effects from the ARMA-filtered residuals, although it does not fully eliminate all residual serial dependence.

Third, the custom EGARCH-X results show that the February 2010 devaluation event is the most important event-control variable in the conditional variance equation. The BIC-selected EGARCH-X specification includes only the devaluation dummy. Richer absolute-shock specifications provide additional evidence that oil shocks, and more weakly DXY shocks, are associated with higher volatility, but these models are not selected by BIC. Therefore, the EGARCH-X evidence supports selective shock transmission rather than broad and uniform global shock effects.

Fourth, the supplementary HAC and weekly SVAR results reinforce the selective interpretation. The HAC absolute-shock model performs better than the signed-shock model and shows that larger U.S. yield, DXY, oil, and VN-Index movements are associated with higher estimated conditional volatility. Weekly SVAR evidence shows that U.S. yield changes help predict USD/VND returns and oil shocks help predict selected volatility, but most other shock variables are insignificant in Granger causality tests.

Overall, USD/VND exchange-rate volatility is best characterized as persistent, fat-tailed, event-sensitive, and selectively exposed to global and domestic shock magnitudes. The evidence is strongest for volatility persistence, heavy tails, the February 2010 devaluation episode, and absolute shock magnitudes. The evidence is weaker for sign asymmetry and for uniform global shock transmission across all variables and models.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

A. Discussion

This study examines the transmission of global and domestic financial shocks to USD/VND exchange rate volatility using daily data from 26 January 2010 to 31 December 2025. The empirical framework combines ARMA mean filtering, GARCH-family volatility models, custom EGARCH-X shock-transmission specifications, HAC shock-volatility regressions, and supplementary weekly SVAR analysis. Overall, the results show that USD/VND exchange rate volatility is persistent, fat-tailed, event-sensitive, and more strongly related to shock magnitudes than to signed shock directions.

The preliminary diagnostic results provide strong support for the use of conditional volatility models. USD/VND returns are stationary after transformation, but they exhibit serial dependence, volatility clustering, conditional heteroskedasticity, and extreme non-normality. The high kurtosis of USD/VND returns indicates that large exchange-rate movements occur much more frequently than under a normal distribution. This justifies the use of Student-(t) innovations in the GARCH-family models. The diagnostic evidence also shows that

modelling the conditional mean alone is insufficient, because ARCH effects remain after ARMA filtering.

The conditional mean results show that ARMA(3,2) is the preferred mean-filtering specification. This model removes most short-run linear dependence in USD/VND returns, but it does not eliminate conditional heteroskedasticity. Therefore, the ARMA model is not treated as the final model. Instead, it is used as a first-stage filter before estimating volatility models. Among the standard GARCH-family specifications, ARMA(3,2)-EGARCH(1,1)-(t) provides the best baseline volatility model. It has the most favourable information criteria among acceptable models and successfully removes remaining ARCH effects from the ARMA-filtered residuals. The significant EGARCH persistence parameter indicates that volatility shocks decay slowly, meaning that periods of exchange-rate risk tend to persist over time.

The custom EGARCH-X results provide more specific evidence on shock transmission. The BIC-selected EGARCH-X specification includes only the 11 February 2010 devaluation dummy. This implies that the strongest improvement to the variance equation comes from controlling for a major exchange-rate policy event rather than from adding a broad set of global and domestic shock variables. However, richer EGARCH-X specifications still provide selective evidence of global shock transmission. In particular, lagged absolute oil-price shocks are statistically significant, while lagged absolute DXY shocks are weakly significant in the specification that includes the devaluation dummy. These results suggest that the magnitude of selected global shocks contains useful information about USD/VND volatility, but the evidence is not uniform across all shock variables.

The asymmetry result should be interpreted cautiously. Although EGARCH-type models allow positive and negative shocks to affect volatility differently, the estimated asymmetry coefficient is not statistically significant in the selected EGARCH-X model. Therefore, the evidence does not support a strong sign-asymmetry interpretation. For USD/VND, the more relevant mechanism appears to be the size of shocks and the presence of discrete policy-related exchange-rate events, rather than whether daily shocks are positive or negative.

The HAC shock-volatility regressions strengthen this interpretation. The absolute-shock model performs better than the signed-shock model, with higher adjusted (R^2) and lower AIC and BIC. In the absolute – shocks specification, lagged absolute movements in U.S. 10 – year Treasury yields, DXY, oil prices, and VN – Index returns are positively and significantly associated with estimated rate volatility. However, the HAC results should be interpreted as reduced form association evidence rather than causal estimates, because the dep

The weekly SVAR results provide supplementary dynamic evidence. Granger causality tests show that U.S. 10-year Treasury yield changes help predict weekly USD/VND returns, while oil-price shocks help predict weekly selected conditional volatility. The impulse response estimates also suggest that VIX and U.S. yield shocks generate short-run responses in USD/VND returns, while VIX and oil shocks are dynamically

related to selected volatility. However, most shock variables are not significant in the weekly Granger tests, and the impulse responses are reported without confidence bands. Therefore, the SVAR evidence should be treated as supportive rather than as the main basis for the study's conclusions.

Taken together, the findings show that USD/VND volatility is shaped by three main forces. First, volatility is highly persistent and strongly driven by past exchange-rate shocks. Second, major exchange-rate policy events, especially the February 2010 devaluation episode, are important volatility events. Third, selected global and domestic shock magnitudes are associated with higher volatility, especially in the HAC evidence. The results therefore support a selective shock-transmission interpretation, not a broad claim that all global shocks uniformly explain USD/VND volatility.

B. Limitations

This study has several limitations. First, the empirical framework uses a two-step ARMA–GARCH structure rather than a fully joint ARMA–GARCH maximum-likelihood model. The ARMA(3,2) model is estimated first, and its residuals are then passed into the GARCH-family models. This approach is transparent and suitable for the study's modelling pipeline, but a fully joint estimation approach could produce more efficient parameter estimates.

Second, the selected conditional volatility used in the HAC and SVAR analyses is a model-generated variable. Although the ARMA(3,2)-EGARCH(1,1)-(t) model passes the core volatility diagnostics, the extracted volatility series is still an estimated measure rather than an observed variable. Therefore, the HAC results should not be interpreted as structural causal estimates.

Third, the custom EGARCH-X result for the 11 February 2010 devaluation dummy reaches an optimizer boundary. This indicates that the event is important for model fit, but its coefficient magnitude should not be interpreted as a precise marginal effect. The devaluation dummy is better understood as an event-control indicator for an exceptional exchange-rate policy episode.

Fourth, the domestic side of the model is represented mainly by VN-Index returns. This captures domestic equity-market conditions but does not fully represent Vietnam's macro-financial environment. Future work could add domestic interest rates, inflation, foreign reserves, capital-flow measures, intervention indicators, or sovereign risk variables if reliable daily or high-frequency data are available.

Fifth, the weekly SVAR analysis depends on the recursive ordering assumption and does not report impulse-response confidence bands. As a result, the SVAR results provide useful dynamic evidence but should not be treated as definitive structural causality. Future research could improve this part by adding bootstrap confidence intervals, forecast error variance decomposition, and alternative identification orderings.

C. Conclusion

This paper investigates shock transmission to USD/VND exchange rate volatility using daily data from 2010 to 2025. The

results show that USD/VND returns exhibit strong volatility clustering, conditional heteroskedasticity, fat-tailed innovations, and persistent conditional volatility. These findings justify the use of ARMA mean filtering and GARCH-family volatility models.

The ARMA(3,2)-EGARCH(1,1)-(t) model is selected as the preferred baseline volatility specification. It provides the best balance between fit and parsimony among the standard GARCH-family models and successfully captures the conditional heteroskedasticity in USD/VND returns. The custom EGARCH-X results show that the February 2010 devaluation episode is an important volatility event, while richer shock specifications provide selective evidence that absolute oil and DXY shocks are associated with higher exchange-rate volatility.

The HAC regressions provide stronger supplementary evidence that shock magnitudes matter more than signed shock directions. Absolute U.S. interest-rate movements, DXY movements, oil-price shocks, and VN-Index movements are positively associated with estimated USD/VND conditional volatility. The weekly SVAR results provide additional but limited dynamic support, showing predictive relevance mainly for U.S. yield shocks in USD/VND returns and oil shocks in selected volatility.

Overall, the study concludes that USD/VND exchange rate volatility is persistent, fat-tailed, sensitive to major policy events, and selectively associated with global and domestic shock magnitudes. The strongest evidence concerns volatility persistence, extreme tail behaviour, and the February 2010 devaluation episode. Evidence for broader global shock transmission exists, but it is selective rather than uniform across all variables and models.

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